```
3000 'FFT FOR REAL SIGNALS
3010 'Upon entry, N% contains the number of points in the DFT, REX[ ] contains
3020 'the real input signal, while values in IMX[ ] are ignored.  Upon return,
3030 'REX[ ] and IMX[ ] contain the DFT output. All signals run from 0 to N%-1.
3040 '
3050 NH% = N%/2-1                          'Separate even and odd points
3060 FOR I% = 0 TO NH%
3070   REX(I%) = REX(2*I%)
3080   IMX(I%) = REX(2*I%+1)
3090 NEXT I%
3100 '
3110 N% = N%/2                             'Calculate N%/2 point FFT
3120 GOSUB 1000                            '(GOSUB 1000 is the FFT in Table 12-3)
3130 N% = N%*2
3140 '
3150 NM1% = N%-1                           'Even/odd frequency domain decomposition
3160 ND2% = N%/2
3170 N4% = N%/4-1
3180 FOR I% = 1 TO N4%
3190   IM%  = ND2%-I%
3200   IP2% = I%+ND2%
3210   IPM% = IM%+ND2%
3220   REX(IP2%) =  (IMX(I%) + IMX(IM%))/2
3230   REX(IPM%) =  REX(IP2%)
3240   IMX(IP2%) = -(REX(I%) - REX(IM%))/2
3250   IMX(IPM%) = -IMX(IP2%)
3260   REX(I%)   =  (REX(I%) + REX(IM%))/2
3270   REX(IM%)  =  REX(I%)
3280   IMX(I%)   =  (IMX(I%) - IMX(IM%))/2
3290   IMX(IM%)  = -IMX(I%)
3300 NEXT I%
3310 REX(N%*3/4) = IMX(N%/4)
3320 REX(ND2%) = IMX(0)
3330 IMX(N%*3/4) = 0
3340 IMX(ND2%) = 0
3350 IMX(N%/4) = 0
3360 IMX(0) = 0
3370 '
3380 PI = 3.14159265                           'Complete the last FFT stage
3390 L% = CINT(LOG(N%)/LOG(2))
3400 LE% = CINT(2^L%)
3410 LE2% = LE%/2
3420 UR = 1
3430 UI = 0
3440 SR =  COS(PI/LE2%)
3450 SI = -SIN(PI/LE2%)
3460 FOR J% = 1 TO LE2%
3470   JM1% = J%-1
3480   FOR I% = JM1% TO NM1% STEP LE%
3490     IP% = I%+LE2%
3500     TR = REX[IP%]*UR - IMX[IP%]*UI
3510     TI = REX[IP%]*UI + IMX[IP%]*UR
3520     REX[IP%] = REX[I%]-TR
3530     IMX[IP%] = IMX[I%]-TI
3540     REX[I%]  = REX[I%]+TR
3550     IMX[I%]  = IMX[I%]+TI
3560   NEXT I%
3570   TR = UR
3580   UR = TR*SR - UI*SI
3590   UI = TR*SI + UI*SR
3600 NEXT J%
3610 RETURN                                TABLE 12-7
```